

A
Project Report
On
"CyberNexus"

(CE451.01 - Major Project)



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Under the Supervision of
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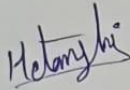
Submitted at



U & P U. PATEL DEPARTMENT OF COMPUTER ENGINEERING
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April 2026

DECLARATION BY THE CANDIDATE

I hereby declare that the project report entitled "CyberNexus" submitted by me to Chandubhai S. Patel Institute of Technology, Changa in partial fulfilment of the requirements for the award of the degree of **B.Tech Computer Engineering**, from U & P U. Patel Department of Computer Engineering, CSPIT, FTE, is a record of bonafide CE451.01 Major Project (project work) carried out by me under the guidance of **Prof. Bijal Suthar**. I further declare that the work carried out and documented in this project report has not been submitted anywhere else either in part or in full and it is the original work, for the award of any other degree or diploma in this institute or any other institute or university.



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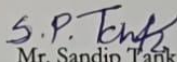
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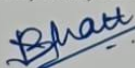
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ABSTRACT

CyberNexus is an AI-powered multi-agent marketplace designed to automate various real-world tasks across domains such as cybersecurity, recruitment, and content analysis. The system integrates multiple intelligent agents into a single platform, where each agent performs a specific function such as email threat detection, resume evaluation, interview scheduling, and presentation generation.

The project utilizes modern AI techniques, including large language models, along with real-time API integrations such as email services, calendar systems, and project management tools. The system follows a modular architecture, allowing easy scalability and addition of new agents. Security is ensured through authentication mechanisms and multi-tenant data isolation.

The main objective of CyberNexus is to reduce manual effort, improve efficiency, and provide a centralized solution for multiple automation tasks. The system demonstrates how AI-driven workflows can be applied to practical scenarios and highlights the potential of agent-based systems in modern software applications.

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I would like to express my sincere gratitude to my project guide Mr. Sandip Tank and Prof. Bijal Suthar for their continuous support, guidance, and valuable suggestions throughout the development of this project. Their insights and feedback helped in improving both the technical and conceptual aspects of the system.

I would also like to thank my organization and team members for providing the opportunity to work on this project and for their support during the development process.

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CHAPTER 1: INTRODUCTION

1.1 PROJECT SUMMARY

CyberNexus is an AI-powered multi-agent marketplace designed to automate a variety of real-world tasks across domains such as cybersecurity, recruitment, and productivity. The system brings together multiple intelligent agents into a single platform, allowing users to perform complex operations with minimal manual effort.

Each agent in CyberNexus is designed to handle a specific task, such as detecting malicious emails, analyzing resumes, scheduling interviews, or generating presentations. These agents work using large language models (LLMs) and are coordinated through a centralized orchestration system. The platform follows a Software-as-a-Service (SaaS) model, ensuring scalability and accessibility for multiple users.

The system also integrates with real-world services such as email platforms, calendars, and project management tools, making it practical and applicable in real scenarios. Security and data isolation are maintained through a multi-tenant architecture, ensuring that user data remains private and secure.

Overall, CyberNexus aims to provide a unified platform where intelligent automation can simplify everyday tasks and improve efficiency.

1.2 PURPOSE

The purpose of CyberNexus is to design and develop a platform that leverages artificial intelligence to automate repetitive and time-consuming tasks. Many processes in areas like cybersecurity monitoring, recruitment, and data analysis still require significant manual effort, which can lead to inefficiencies and human errors.

CyberNexus addresses this challenge by introducing a system where multiple AI agents can independently and collaboratively perform these tasks. The goal is to reduce manual workload, improve accuracy, and enable faster decision-making.

Additionally, the project aims to explore the practical implementation of agent-based AI systems, real-time integrations with external services, and secure multi-user system design. It serves both as a functional solution and as a learning platform for understanding modern AI-driven software architectures.

1.3 OBJECTIVE

The objectives of this system are as follows:

- To develop an AI-powered multi-agent marketplace for automating real-world tasks
- To design and implement intelligent agents for cybersecurity, recruitment, and productivity use cases
- To integrate real-time external APIs such as Gmail, Google Calendar, Jira, and VirusTotal
- To implement LLM-based decision-making using modern AI frameworks
- To build a secure multi-tenant system ensuring user data isolation
- To automate workflows such as email threat detection, interview scheduling, and resume evaluation
- To provide a centralized and user-friendly platform for accessing multiple AI services
- To improve overall efficiency, accuracy, and productivity through automation

1.4 SCOPE

The scope of CyberNexus includes the design, development, and deployment of a multi-agent AI platform that provides automation services through various intelligent agents.

What the system does:

- Provides a centralized platform to access multiple AI agents
- Automates cybersecurity tasks such as email threat detection and header validation

- Supports recruitment workflows including resume analysis and interview scheduling
- Performs content analysis and compliance verification
- Generates presentations from structured data or user input
- Integrates with external APIs for real-time data processing
- Ensures secure user authentication and data isolation

What the system does not do:

- It does not replace full enterprise systems like complete HR or cybersecurity platforms
- It does not guarantee 100% accuracy in AI-generated outputs due to limitations of LLMs
- It does not control or manage third-party APIs beyond integration usage
- It does not provide offline functionality, as it depends on cloud-based services

1.5 TECHNOLOGY AND LITERATURE REVIEW

➤ Technology Stack:

Table 1.1 Technologies Used and Its Purpose

Technology	Purpose
Python	Backend development and implementation of business logic
LangChain, LangGraph	Building and managing AI agent workflows
PostgreSQL	Database management and storage of application data
Docker	Containerization and deployment of the application
Postman	Testing APIs and validating backend services

➤ Literature Review:

The concept of AI agent marketplaces and automation platforms has been explored by several industry solutions such as Kore.ai and Druid AI. These platforms provide enterprise-level AI solutions focused on conversational interfaces, automation, and workflow management.

Kore.ai offers a comprehensive platform for building virtual assistants and enterprise automation systems. It provides advanced tools for conversation design, integration with enterprise systems, and scalable deployment. However, its primary focus is on chatbot-based automation and enterprise workflows rather than modular multi-agent systems.

Similarly, Druid AI focuses on conversational AI solutions that enable businesses to automate processes through virtual assistants. It provides integration capabilities with enterprise applications and emphasizes user interaction through conversational interfaces.

In comparison, CyberNexus differs in its approach by focusing on a multi-agent architecture, where each agent is specialized for a specific task and operates independently within a unified system. Instead of relying solely on conversational interfaces, CyberNexus emphasizes task-oriented automation using LLM-driven workflows.

Another key difference is the integration of diverse functionalities such as cybersecurity analysis, recruitment automation, and presentation generation within a single platform. This makes CyberNexus more versatile for academic and experimental purposes.

Additionally, CyberNexus incorporates modern AI frameworks and event-driven architectures to enable real-time processing and intelligent decision-making. While platforms like Kore.ai and Druid AI are designed for large-scale enterprise use, CyberNexus demonstrates a more flexible and modular approach suitable for experimentation, learning, and targeted automation.

CHAPTER 2: PROJECT MANAGEMENT

2.1 PROJECT PLANNING

2.1.1 Project Development Approach and Justification

The Agile development model was adopted for this project. This approach was chosen because the project requirements were evolving as new ideas and improvements were identified during development.

Instead of building the entire system at once, the project was developed incrementally by implementing one agent (use case) at a time. After completing each agent, testing and evaluation were carried out before proceeding further.

The Agile model was suitable for this project due to the following reasons:

- It allows flexibility in design and implementation
- Supports continuous testing and improvement
- Helps in handling changing requirements and new ideas
- Enables faster feedback and iteration
- Suitable for AI-based systems where outputs need refinement

This approach ensured that each component of CyberNexus was properly validated before integration into the overall system.

2.1.2 Project Effort and Time, Cost Estimation

The effort and time required for developing CyberNexus were estimated using the COCOMO (Constructive Cost Model).

Since this project falls under a moderate complexity (semi-detached type), the estimation considers factors such as AI integration, API usage, and multi-agent architecture.

Estimation Parameters (Approximate)

- Estimated Lines of Code (LOC): ~25,000

- Development Type: Semi-detached

Effort Equation:

$$\text{Effort} = a \times (\text{KLOC})^b$$

Using standard COCOMO constants:

- $\text{Effort} \approx 2.8 \times (25)^{1.05} \approx \sim 80$ person-months (approx.)

Time Estimation:

- $\text{Development Time} \approx 2.5 \times (\text{Effort})^{0.38}$
- $\approx \sim 12\text{--}14$ months (ideal full-scale project)

However, for academic implementation, the project was completed in a shorter duration using focused development and modular implementation.

Cost Estimation:

Since the project is developed in an academic environment:

- No direct financial cost involved
- Tools and frameworks used are mostly open-source or free
- Cost mainly involves development effort and time

2.1.3 Roles and Responsibilities

The project was developed by interns with clearly defined roles:

Frontend Developer (1 Intern)

- Responsible for designing and developing the user interface
- Implemented dashboards and interaction components
- Ensured smooth user experience

Backend Developers (1 Interns – AI Developer)

- Designed and implemented all AI agents
- Developed:
 - o Email Security Agent
 - o Resume Reviewer Agent
 - o Scheduler Agent
 - o Content Reviewer Agent
 - o Compliance Agent
 - o AutoPresenter & DB Presenter
 - o Newsletter Agent
- Integrated LLM workflows using LangChain and LangGraph
- Handled API integrations and core logic

2.1.4 Group Dependencies

The successful development of CyberNexus required coordination between different team members and components.

- Frontend depended on backend APIs for displaying results
- Backend depended on AI models and external APIs for processing
- AI agents depended on proper input validation from frontend
- Integration tasks required synchronization between all team members

Additionally:

- External APIs (Gmail, Jira, etc.) were critical dependencies
- Any delay or failure in API responses affected system workflows

Regular communication and updates were maintained to ensure smooth collaboration.

2.2 PROJECT SCHEDULING

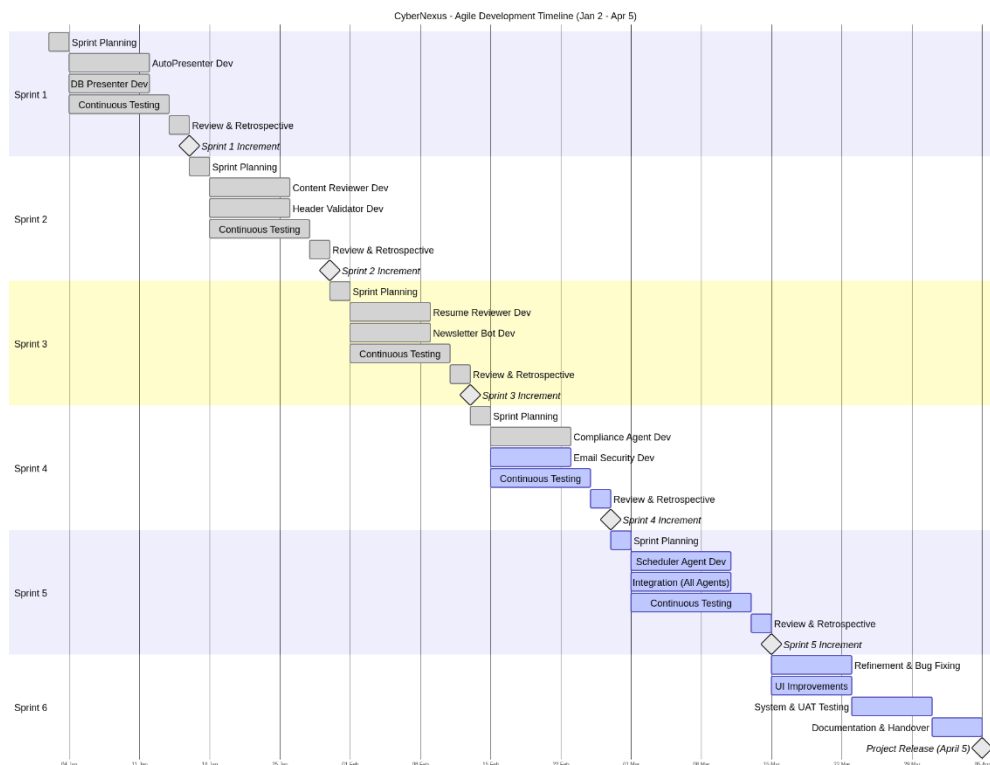


Fig 2.1 Gantt Chart

CHAPTER 3: SYSTEM REQUIREMENTS STUDY

3.1 USER CHARACTERISTICS

CyberNexus is designed to be used by a diverse group of users with varying levels of technical expertise. The system is intended to be simple enough for non-technical users while still providing advanced features for technical users.

1. General Users

These users interact with the platform to perform tasks such as generating presentations, analyzing content, or checking emails. They may not have a strong technical background, so the system interface is designed to be intuitive and easy to use.

2. Recruiters / HR Professionals

Recruiters use the system for resume analysis and interview scheduling. They require reliable and accurate results, especially when evaluating candidates or managing interview slots.

3. Security Analysts

These users primarily use features such as the Email Security Agent and Header Validator Agent. They are expected to have some knowledge of cybersecurity concepts and rely on the system to identify potential threats and vulnerabilities.

4. Technical Users / Developers

These users have a higher level of technical expertise and may use advanced features such as the DB Presenter Agent or API-based integrations. They are comfortable working with structured data and understanding system workflows.

3.2 HARDWARE AND SOFTWARE REQUIREMENTS

CyberNexus is a web-based application and does not require specialized hardware. It is designed to run on standard computing systems with internet connectivity.

Hardware Requirements

- Processor: Minimum Intel i3 or equivalent
- RAM: Minimum 4 GB (8 GB recommended for better performance)
- Storage: At least 1 GB free space
- Device: Desktop or laptop

Software Requirements

- Operating System: Windows
- Web Browser: Google Chrome
- Backend Technologies: Python
- AI Frameworks: LangChain, LangGraph
- Database: PostgreSQL
- Authentication: Firebase Authentication
- Deployment: Docker (optional but recommended)

External Tools and APIs

- Gmail API (for email processing)
- Google Calendar API (for scheduling)
- Jira API (for ticket analysis)
- VirusTotal API (for threat detection)

3.3 ASSUMPTIONS AND DEPENDENCIES

➤ Assumptions:

- Users have a stable internet connection
- External APIs are accessible and functioning properly
- Input data provided by users (emails, resumes, URLs) is valid
- AI models provide reasonably accurate outputs
- Users follow proper usage guidelines

➤ Dependencies:

- External APIs:
 - Gmail API → Email monitoring and processing
 - Google Calendar API → Scheduling functionality
 - Jira API → Ticket retrieval and analysis
 - VirusTotal API → Security scanning
- AI Frameworks:
 - LangChain and LangGraph for agent workflows

- LLM services for reasoning and decision-making
- Database System:
 - PostgreSQL for storing reports and user-related data
 - Row-Level Security (RLS) for data isolation
- Authentication System:
- Firebase Authentication for secure user login

CHAPTER 4: SYSTEM ANALYSIS

4.1 STUDY OF CURRENT SYSTEM

In many organizations, tasks related to cybersecurity, recruitment, and content analysis are still performed manually or with the help of isolated tools. For example, email security often relies on basic spam filters, which are not always effective in detecting advanced phishing or fraud attempts. Similarly, recruitment processes such as resume screening and interview scheduling are time-consuming and require significant human effort.

Existing systems typically focus on solving a single problem. For instance, there are separate tools for email filtering, resume screening, and content checking, but they do not work together as a unified system. This results in inefficiencies, duplication of effort, and lack of coordination between processes.

Moreover, many of these tools lack intelligent decision-making capabilities. They depend on predefined rules rather than adaptive learning or contextual understanding. As a result, they may fail to handle complex or dynamic scenarios effectively.

Overall, the current system environment is fragmented, less efficient, and highly dependent on manual intervention.

4.2 PROBLEM AND WEAKNESSES OF CURRENT SYSTEM

- Heavy dependence on manual effort, leading to slower processing
- Lack of integration between different tools and systems
- Limited ability to handle complex or dynamic scenarios
- Traditional systems rely on rule-based logic, which is less flexible
- Higher chances of human error in tasks like resume screening or threat detection
- No centralized platform to manage multiple workflows
- Inefficient handling of large volumes of data
- Lack of real-time processing and automation

4.3 REQUIREMENTS OF NEW SYSTEM

To overcome the limitations of the current system, a new solution is required that integrates automation, intelligence, and scalability into a single platform.

The new system, CyberNexus, is designed to:

- Provide a centralized platform for multiple tasks
- Use AI-based agents for intelligent decision-making
- Automate workflows to reduce manual effort
- Integrate with real-world systems through APIs
- Ensure secure and scalable architecture

4.3.1 Functional Requirements

- The system shall allow users to log in securely using authentication mechanisms
- The system shall provide a marketplace interface to select and execute AI agents
- The system shall support execution of multiple agents such as:
 - o Email Security Agent
 - o Resume Reviewer Agent
 - o Scheduler Agent
 - o Content Reviewer Agent
 - o Compliance Agent
 - o AutoPresenter and DB Presenter
 - o Newsletter Agent
- The system shall process user input and generate appropriate outputs using AI models
- The system shall integrate with external APIs to:
 - o Fetch emails
 - o Access calendar data
 - o Retrieve Jira tickets
 - o Scan links
- The system shall allow users to view and download results such as reports and presentations
- The system shall ensure data isolation using tenant-based architecture

- The system shall perform real-time processing for certain workflows (e.g., email monitoring)

4.3.2 Non-Functional Requirements

➤ Performance:

- AI agents are executed asynchronously, which allows multiple requests to be processed without blocking the system
- Real-time workflows such as email monitoring are handled using webhook-based triggers instead of continuous polling
- Database operations are optimized using JSONB storage and UPSERT queries, reducing redundant processing
- Only necessary data is stored in the database (e.g., compliance reports), avoiding unnecessary load

➤ Reliability:

- External API failures (e.g., Gmail or VirusTotal) are handled using error handling and fallback mechanisms
- The system does not depend on a single component; agents operate independently, reducing failure impact
- Critical data such as compliance reports is stored in the database to prevent data loss
- Logging mechanisms are used to track system behavior and debug issues

➤ Security:

- The system requires user authentication using Firebase Authentication before executing any agent
- All requests are validated using secure tokens, preventing unauthorized access
- A multi-tenant architecture is implemented using `tenant_id`, ensuring that users can only access their own data
- Sensitive operations such as email processing and API access are performed through OAuth-secured APIs

➤ Scalability:

- The application is containerized using Docker, allowing easy scaling and deployment
- New AI agents can be added without modifying existing components due to modular architecture
- Database design using `tenant_id` supports multiple users without data conflicts

➤ **Usability:**

- A web-based interface is provided where users can select agents and provide input easily
- Each agent follows a simple workflow: input → process → output
- Outputs such as reports and presentations are displayed in a structured and readable format

4.4 FEASIBILITY STUDY

4.4.1 Does the system contribute to the overall objectives of the organization?

Yes, CyberNexus contributes positively to organizational goals by improving efficiency, reducing manual workload, and enhancing decision-making through automation.

4.4.2 Can the system be implemented using the current technology and within the given cost and schedule constraints?

Yes, CyberNexus can be implemented using currently available technologies and within reasonable cost and time constraints.

4.4.3 Can the system be integrated with other systems which are already in place?

Yes, the system can be integrated and worked with existing systems.

4.5 ACTIVITY/PROCESS IN NEW SYSTEM

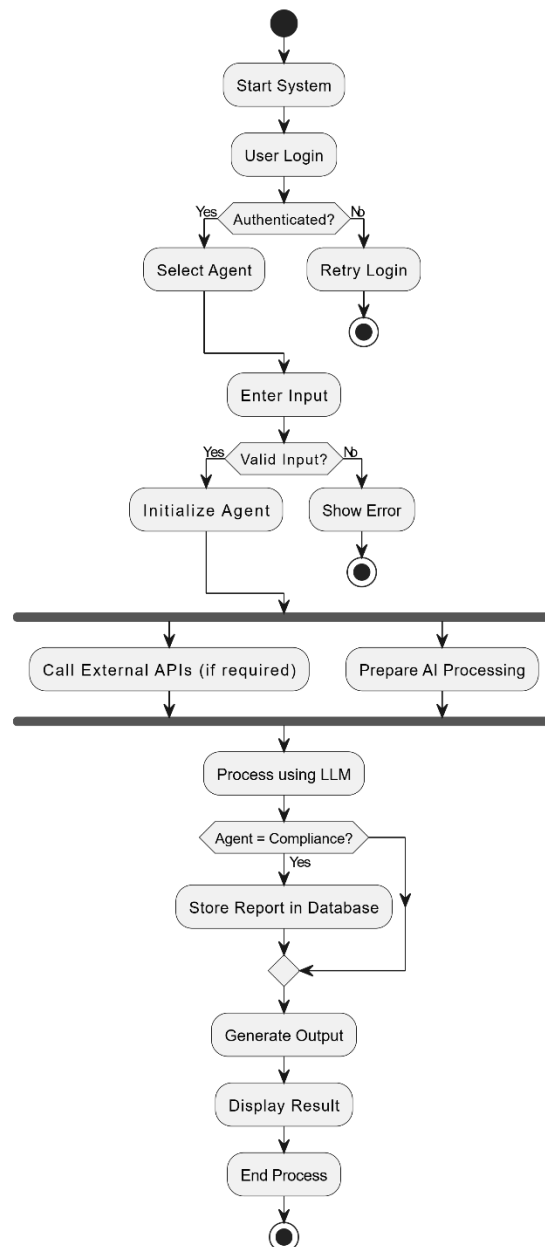


Fig 4.1 Activity Diagram

4.6 FEATURE OF NEW SYSTEM

- AutoPresenter Agent
Generates PowerPoint presentations automatically from a given topic or structured JSON input, including content, charts, and formatted slides
- DB Presenter Agent
Converts natural language queries into SQL, fetches data from PostgreSQL, and generates presentation-ready outputs with visualizations
- Content Reviewer Agent

Crawls a given website and analyzes content to detect spelling, grammar, punctuation, and typographical errors

- Header Validator Agent

Performs security analysis of HTTP headers and identifies missing or misconfigured security settings in websites

- Resume Reviewer Agent

Evaluates candidate resumes against a job description, assigns scores, extracts key details, and generates ranking reports

- Newsletter Bot

Fetches recent cybersecurity news, summarizes it using AI, and generates a structured newsletter for posting

- Compliance Verification Agent

Retrieves Jira tickets, analyzes them for compliance alignment, and stores structured reports in the database

- Email Security Agent

Monitors incoming emails in real-time, analyzes content using AI and threat intelligence APIs, and classifies emails as safe, spam, or fraud

- Scheduler Agent

Automates interview scheduling by checking calendar availability, processing candidate responses, and creating meeting events

4.7 USE CASE DIAGRAM

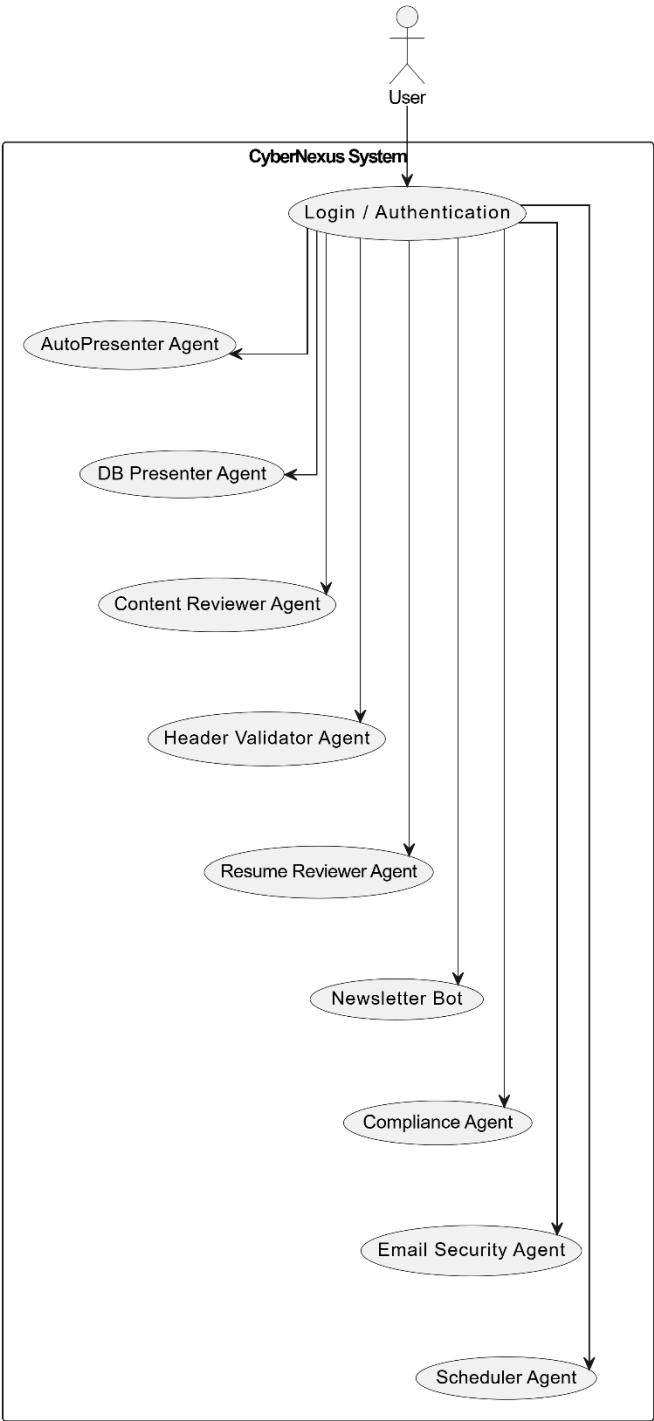
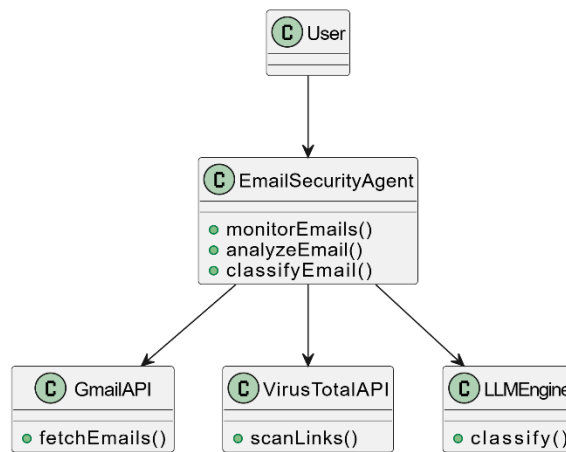
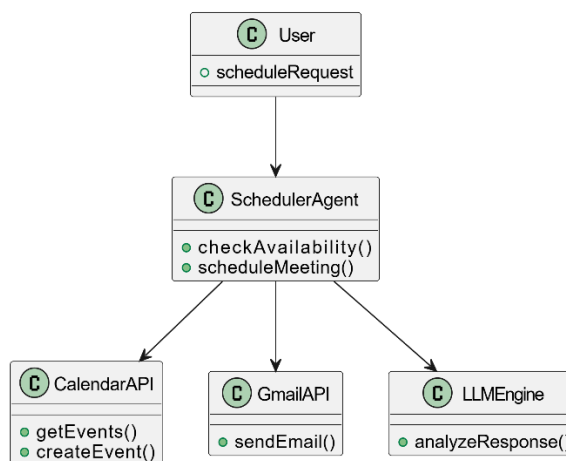
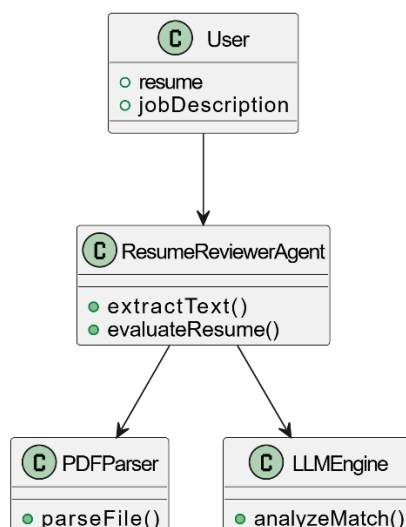
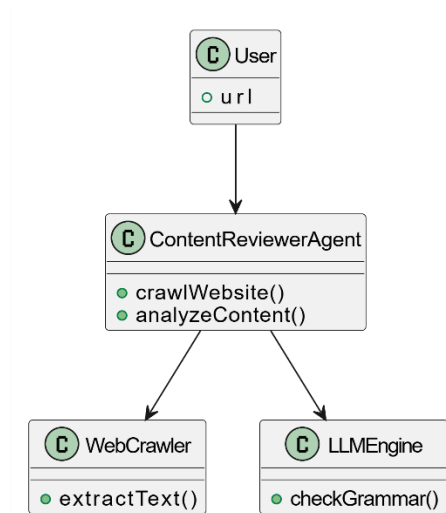
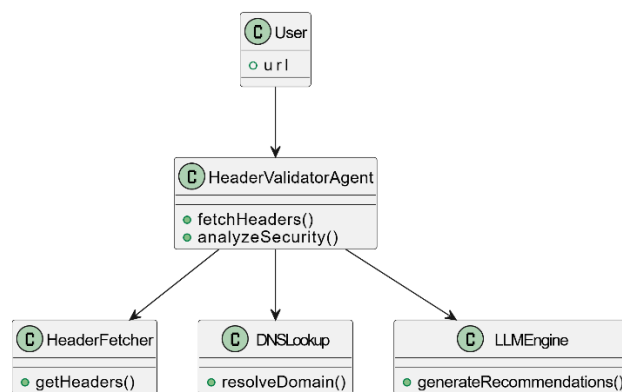
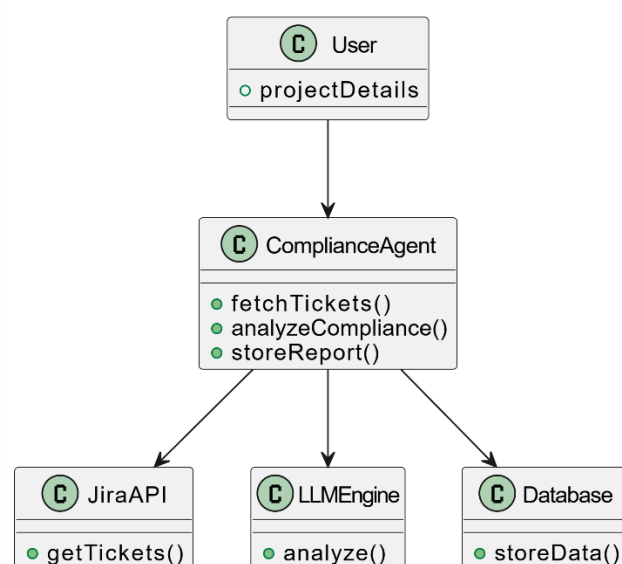
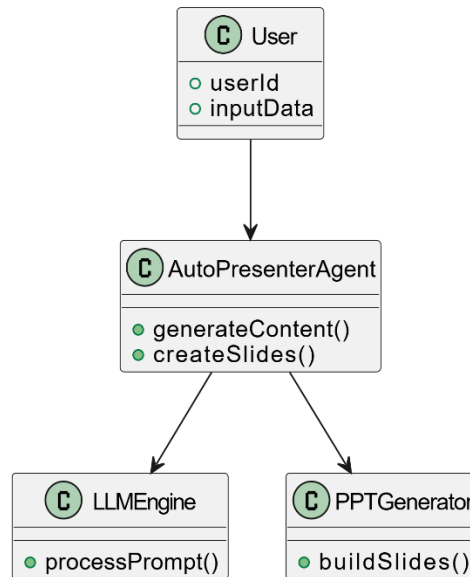
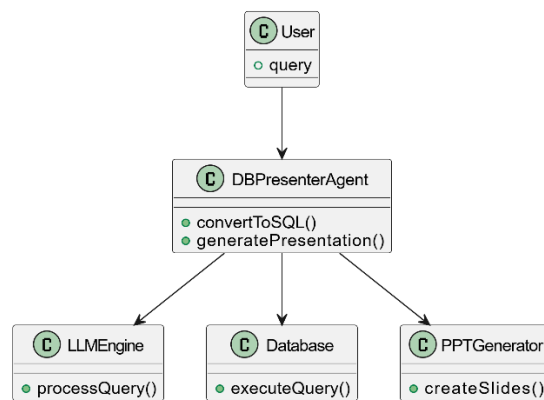
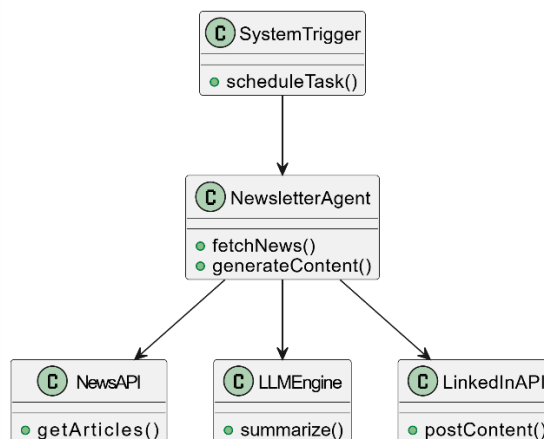


Fig 4.2 Use case Diagram

4.8 CLASS DIAGRAM

**Fig 4.3 Email Security Agent****Fig 4.4 Interview Scheduler Agent****Fig 4.5 Resume Reviewer Agent**

**Fig 4.6 Content Reviewer Agent****Fig 4.7 Header Validator Agent****Fig 4.8 Compliance Verification Agent**

**Fig 4.9 Auto Presenter Agent****Fig 4.10 DB Presenter Agent****Fig 4.11 Cyber Security Newsletter Bot**

4.9 SEQUENCE DIAGRAM

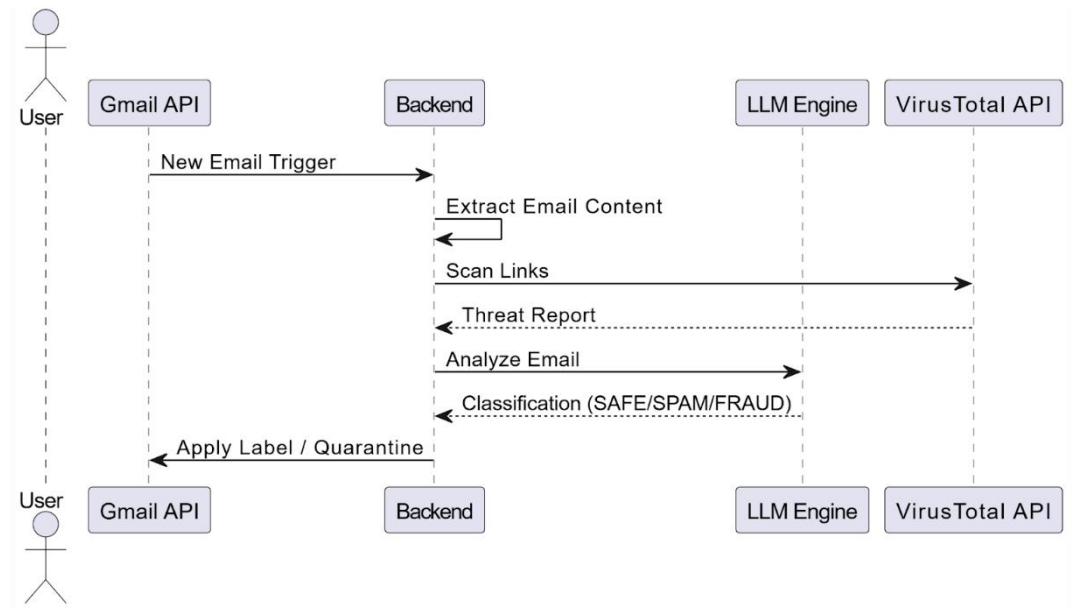


Fig 4.12 Email Security Agent

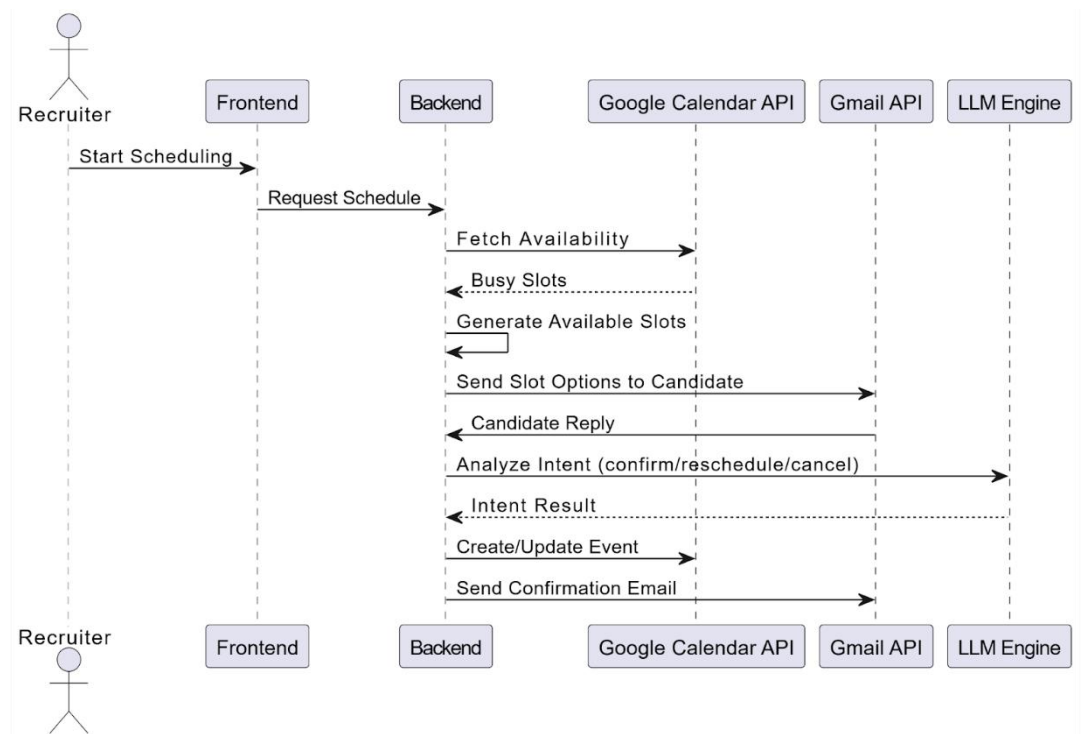


Fig 4.13 Interview Scheduler Agent

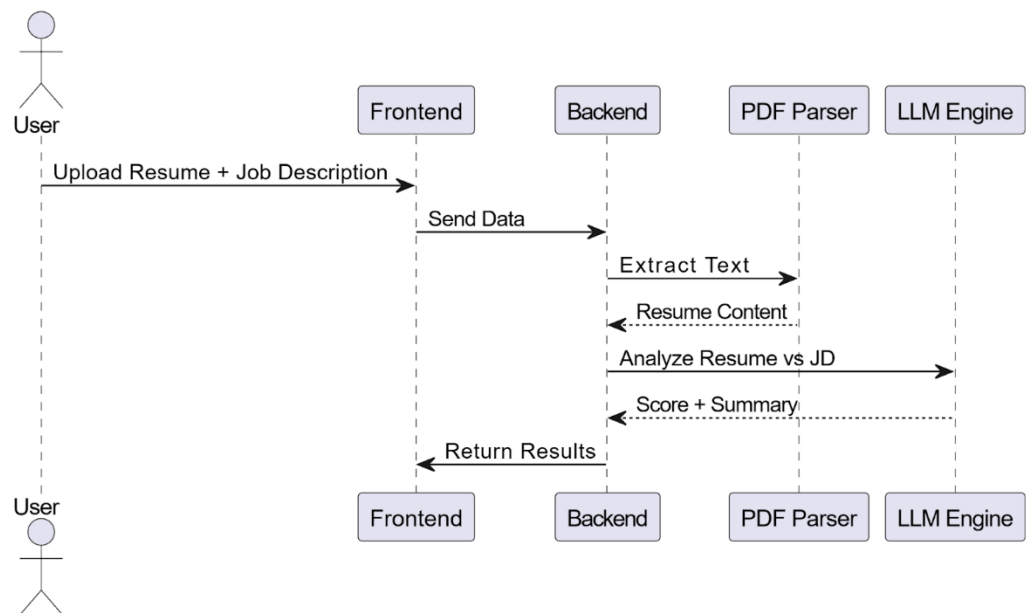


Fig 4.14 Resume Reviewer Agent

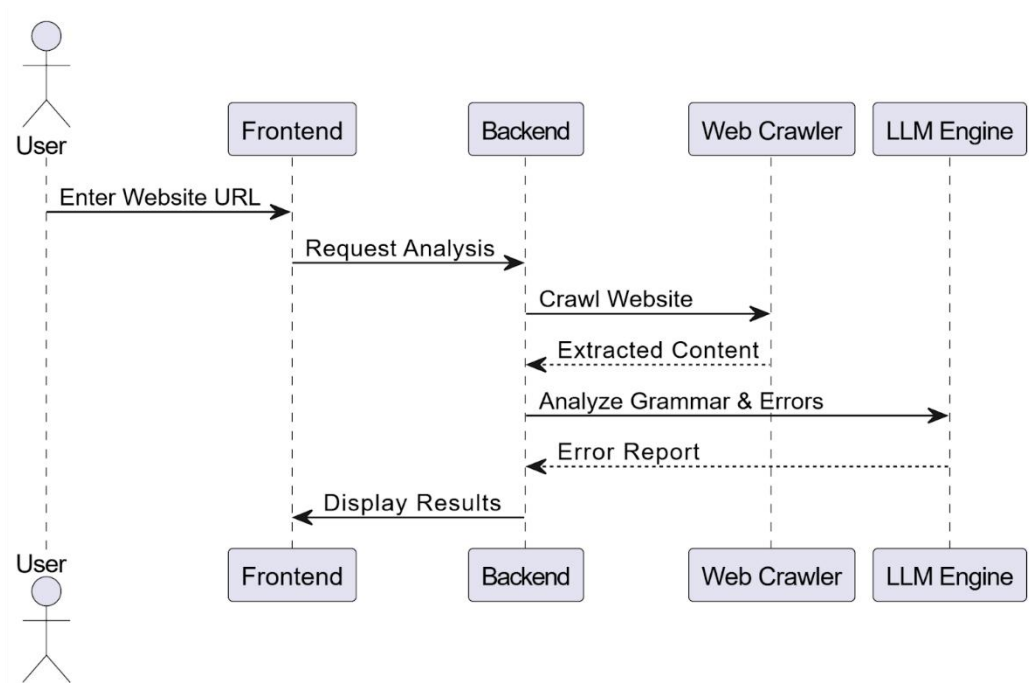


Fig 4.15 Content Reviewer Agent

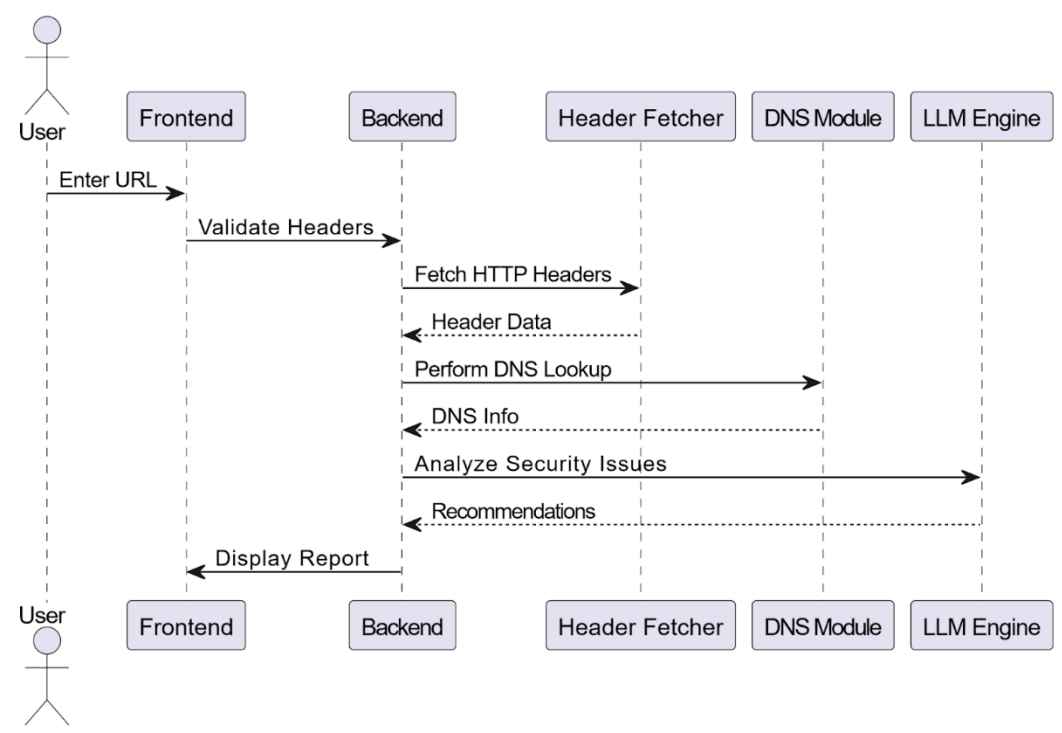


Fig 4.16 Header Validator Agent

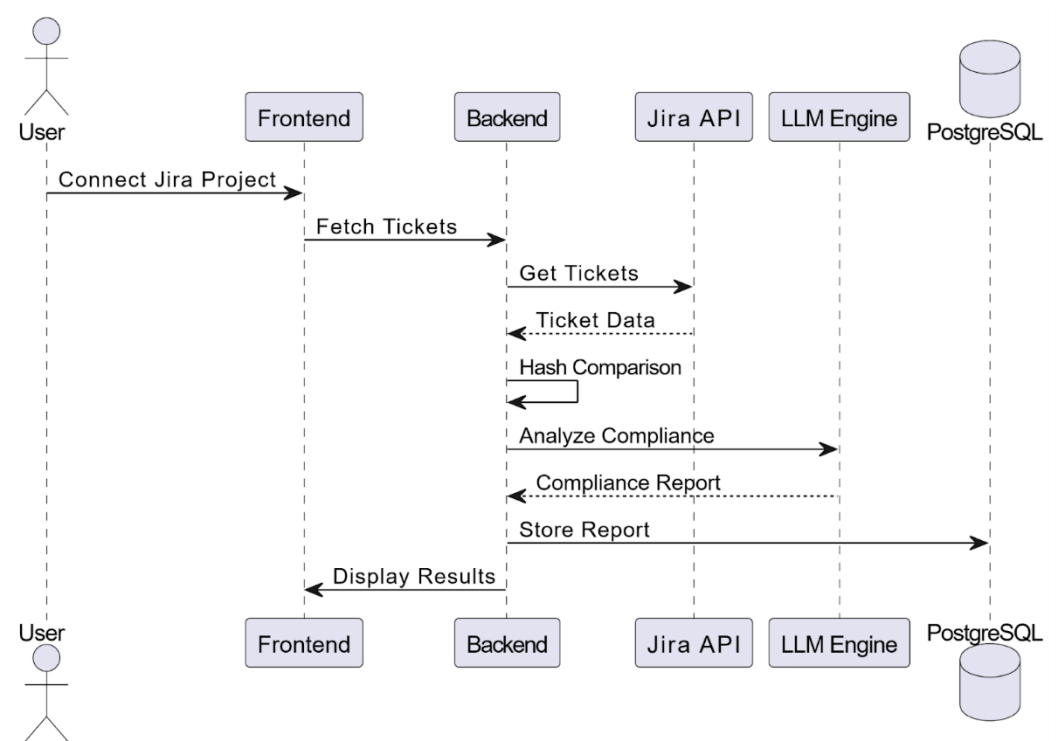


Fig 4.17 Compliance Verification Agent

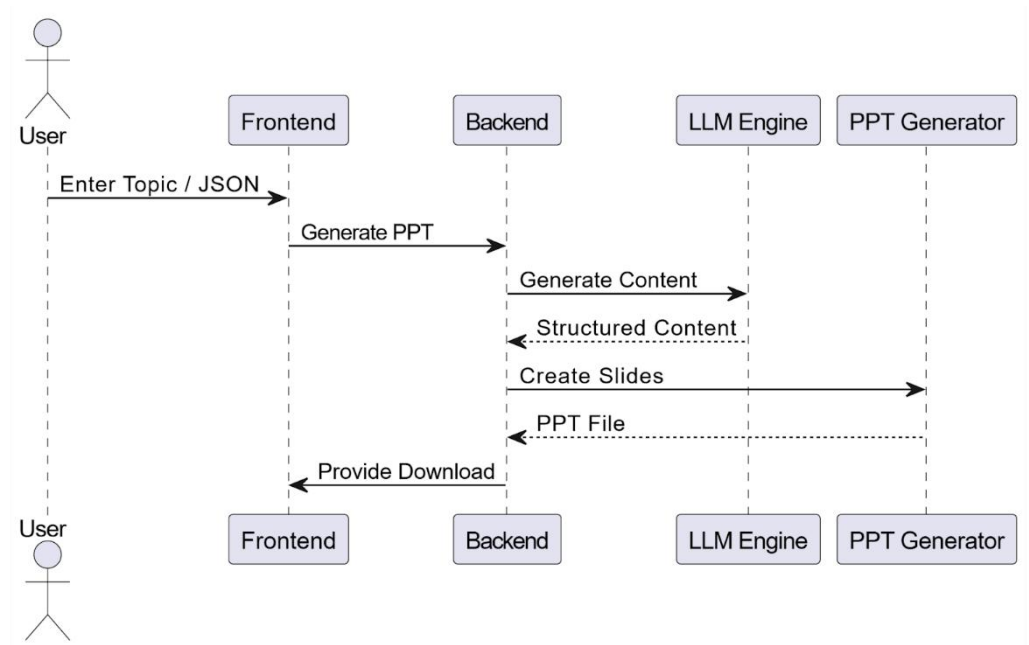


Fig 4.18 Auto Presenter Agent

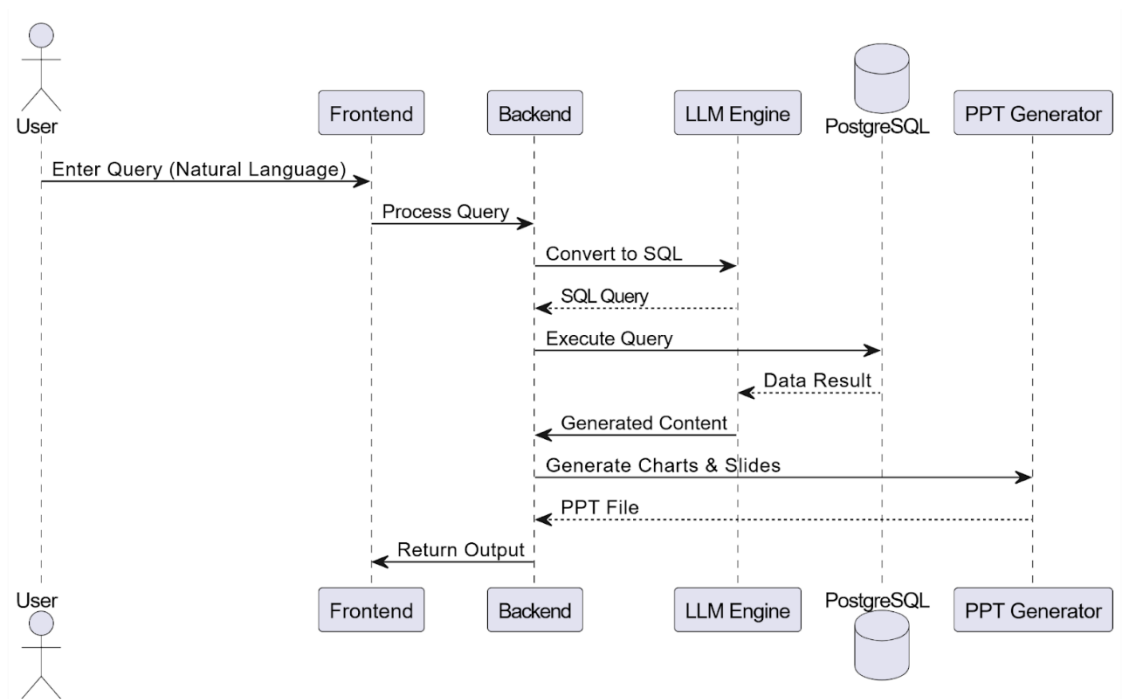


Fig 4.19 DB Presenter Agent

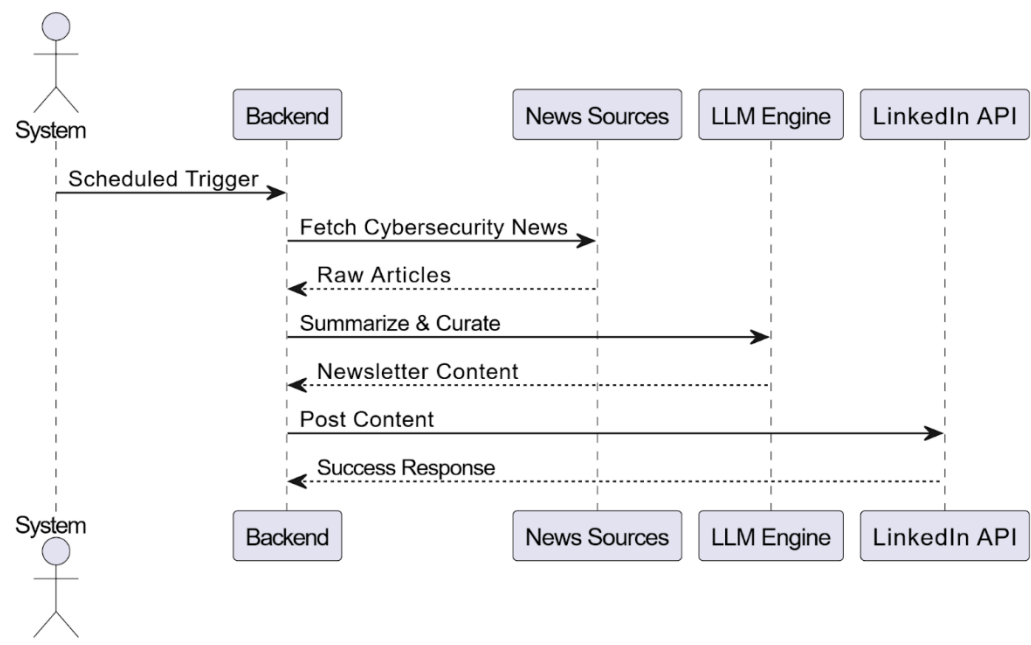


Fig 4.20 Cyber Security Newsletter Bot

CHAPTER 5: SYSTEM DESIGN

5.1 SYSTEM APPLICATION DESIGN

The CyberNexus system is designed using a modular and agent-based architecture. Each functionality in the system is implemented as an independent AI agent, which is responsible for handling a specific task. These agents are coordinated through a centralized backend system that manages request flow, processing, and response generation.

The system follows a layered structure consisting of:

- Frontend layer for user interaction
- Backend layer for request handling and orchestration
- AI layer for intelligent processing using LLMs
- Integration layer for external APIs
- Database layer for storing required data

When a user interacts with the system, the request is first authenticated and then routed to the appropriate agent. The agent processes the input using AI models and external services, and the result is returned to the user.

5.1.1 Method Pseudo Code

Below is a generalized pseudo code representing how an agent in CyberNexus operates:

➤ **User Login:**

START

Receive user request (input data)

Authenticate user

IF authentication fails THEN

Return error message

STOP

END IF

Validate input data

IF input is invalid THEN

Return validation error

STOP

END IF

Identify selected agent

Initialize agent workflow

IF external API is required THEN

Fetch required data from API

END IF

Process input using AI model (LLM)

IF database storage is required THEN

Store processed result in database

END IF

Generate final output

Return result to user

END

5.2 INPUT/OUTPUT AND INTERFACE DESIGN

5.2.1 Samples of Forms, Reports and Interface

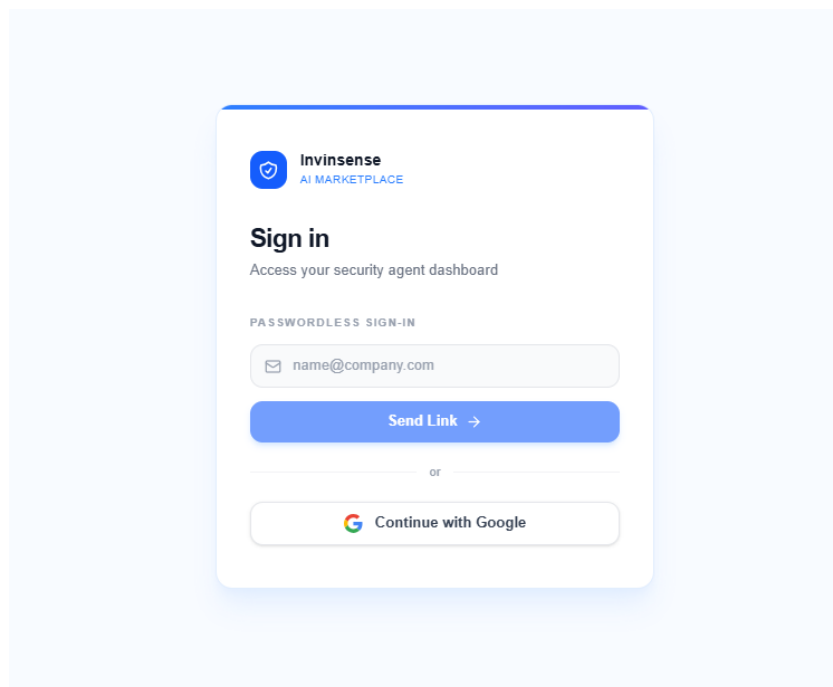


Fig 5.1 Login Form

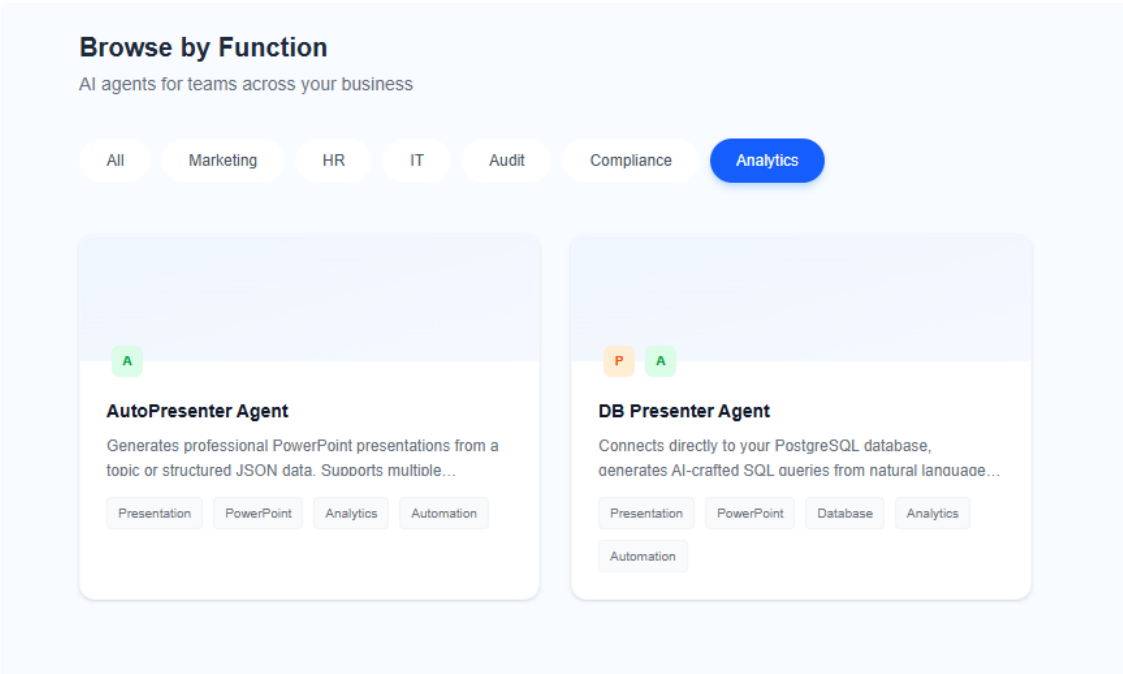


Fig 5.2 Agent Filter

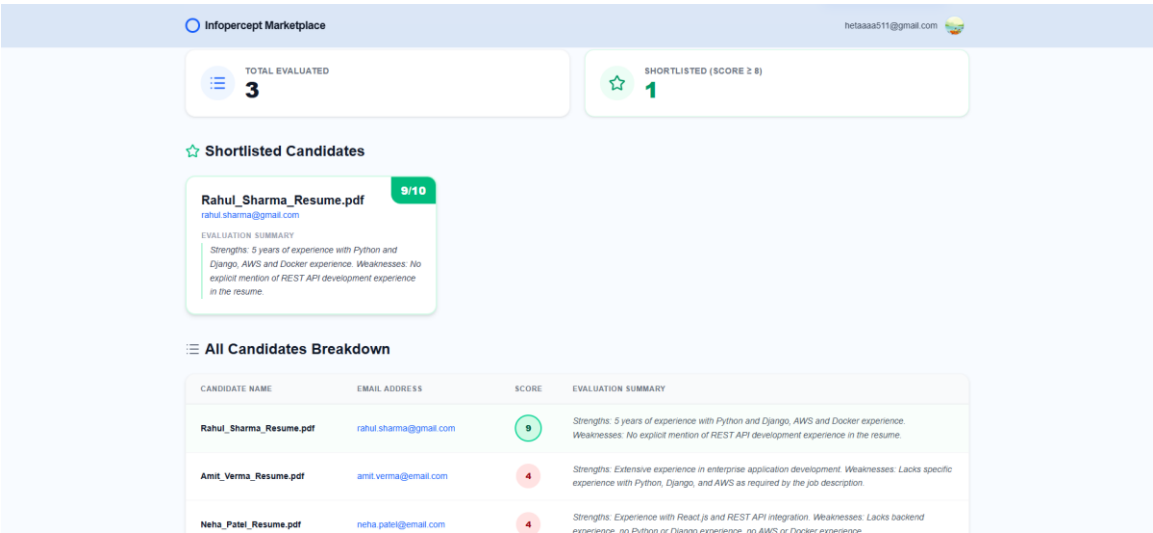


Fig 5.3 Resume Reviewer Report

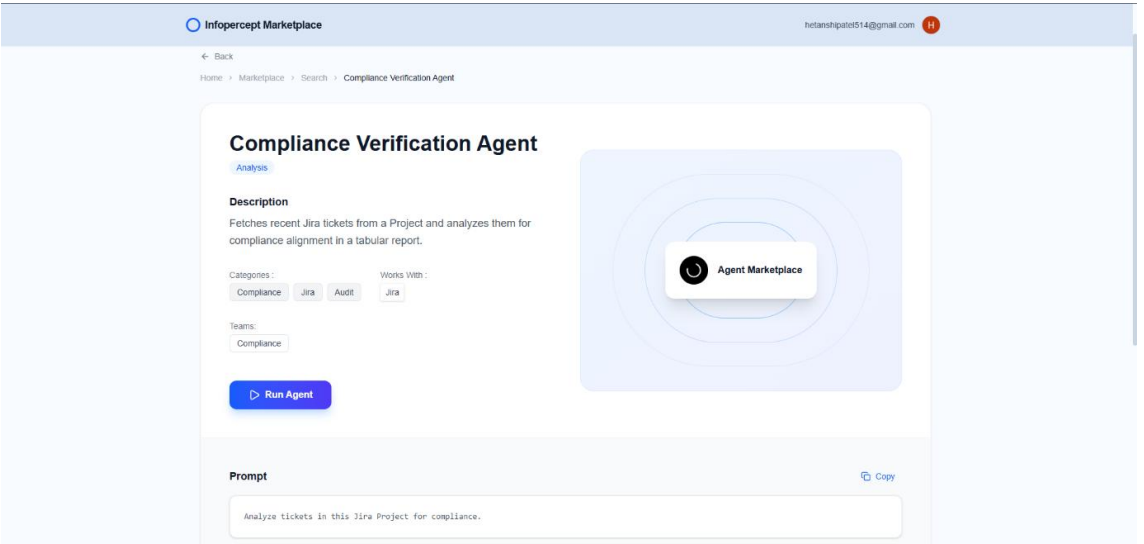


Fig 5.4 Agent Details with Input form

The screenshot displays the 'Compliance Audit Report' in the Infopercept Marketplace. The report title is 'Compliance Audit Report' with a subtitle 'Project Level Analysis • 2/23/2026'. Action buttons for 'Print / PDF' and 'Export CSV' are visible. The report is divided into two main sections: '1. Satisfied Tickets' (green header) and '2. Dissatisfied Tickets' (red header). The 'Satisfied Tickets' section shows a single ticket, CM-4, with a status of 'Done'. The 'Dissatisfied Tickets' section contains a table with the following data:

TICKET	SUMMARY	STATUS	SEVERITY	ALIGNMENT	COMP. %	DEFICIENCIES & GAPS	CHECKLIST
CM-3	Validate Approval Workflow Against Change Management Policy	Done	Medium	Misaligned	50%	MISSING / INCOMPLETE FIELDS: - Missing Approver, Reviewer - Checklist Fail (Approval and Authorization) Approval Workflow: Examine the chain of approvals, ensuring involvement from key stakeholders like IT operations, security team, change advisory board (CAB), and business owners. Check for segregation of duties (e.g., requester ≠ approver ≠ implementer). - Checklist Fail (Post-implementation Review and Closure) Closure Documentation: Confirm the ticket was properly closed with final approvals and that all fields are complete. RECOMMENDED ACTIONS: - Assign an approver and reviewer to the ticket. - Ensure all compliance checklist items are addressed.	VIEW CHECKLIST
CM-6	Implement Multi-Factor Authentication (MFA) for VPN	To Do	High	Misaligned	0%	MISSING / INCOMPLETE FIELDS: - Missing Approver, Reviewer - Checklist Fail (Phone Factorization and Authentication) Detailed Factorization	VIEW CHECKLIST

Fig 5.5 Jira ticket report with all compliance checks

5.2.2 State Diagram

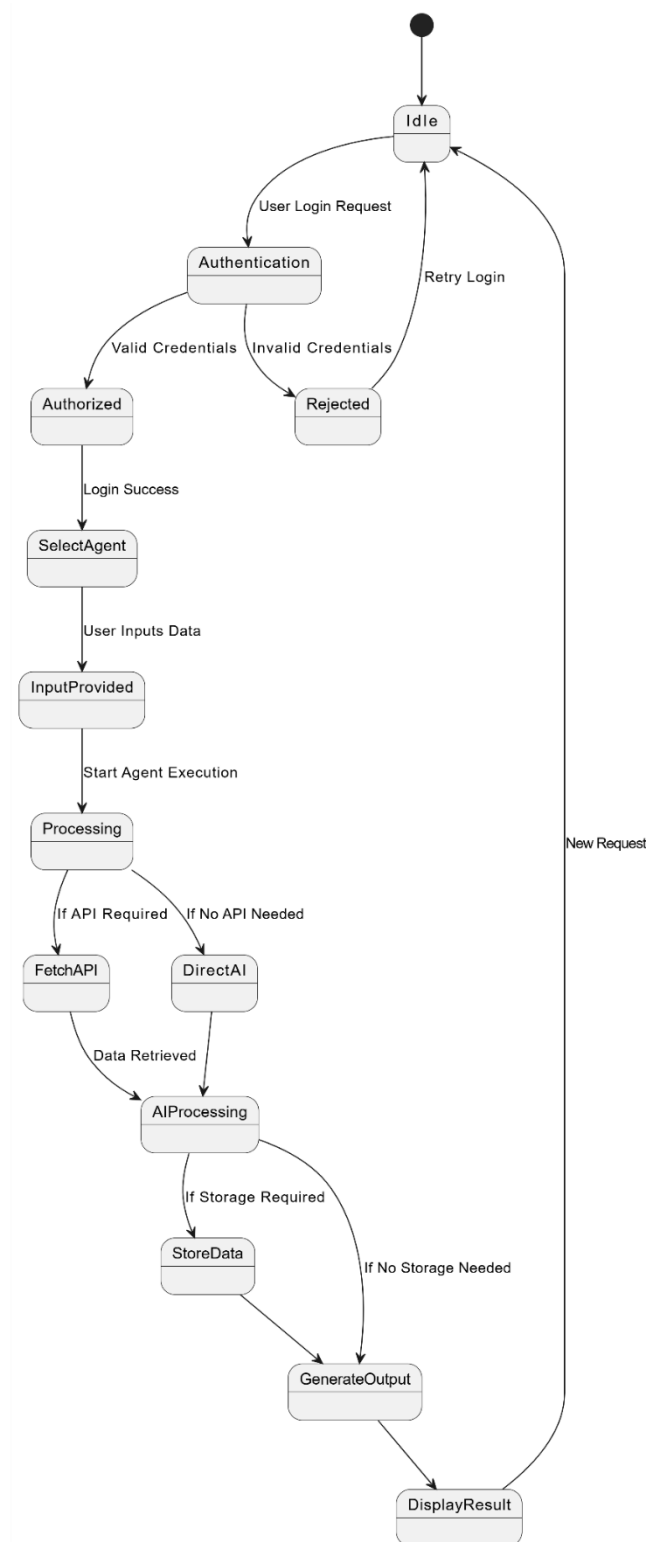


Fig 5.6 State Diagram of the Cyber Nexus

CHAPTER 6: IMPLEMENTATION PLANNING

6.1 IMPLEMENTATION ENVIRONMENT

- Processor: Intel Core i5
- RAM: 8 GB
- Storage: 512 GB SSD
- Internet Connection: Required for API integration and testing
- Operating System: Windows 10
- Code Editor: Visual Studio Code
- Backend Language: Python
- AI Frameworks: LangChain, LangGraph
- Database: PostgreSQL
- Authentication: Firebase Authentication
- API Testing Tool: Postman
- Browser: Google Chrome
- Containerization: Docker (used for deployment/testing)

6.2 MODULES SPECIFICATION

1. Authentication Module

- Handles user login using Firebase Authentication
- Verifies user identity using tokens
- Restricts access to system features without login

2. AutoPresenter Module

- Accepts topic or structured input
- Generates presentation content using AI
- Creates slides using a PPT generation module

3. DB Presenter Module

- Converts natural language input into SQL queries
- Fetches data from PostgreSQL database
- Generates presentation output based on retrieved data

4. Content Reviewer Module

- Accepts website URL as input
- Crawls web pages and extracts content
- Uses AI to detect grammar and spelling errors

5. Header Validator Module

- Fetches HTTP headers from given URL

- Performs DNS lookup
 - Analyzes security-related header configurations
6. Resume Reviewer Module
- Accepts resume and job description
 - Extracts text from resume
 - Uses AI to evaluate and score candidate suitability
7. Newsletter Bot Module
- Fetches cybersecurity news from external sources
 - Summarizes content using AI
 - Generates formatted newsletter output
8. Compliance Verification Module
- Connects to Jira API
 - Fetches project tickets
 - Analyzes compliance and stores reports in database
9. Email Security Module
- Monitors Gmail inbox using API
 - Scans links using VirusTotal
 - Uses AI to classify emails (safe/spam/fraud)
10. Scheduler Module
- Handles Fetches calendar availability using Google Calendar API
 - Processes candidate responses
 - Schedules meetings and sends confirmation emails

6.3 Security Features

The following security features are implemented:

- User authentication is enforced using Firebase Authentication, and no agent can be executed without successful login
- Secure token-based validation is used to verify each request between frontend and backend
- A multi-tenant architecture is implemented using `tenant_id`, ensuring users can only access their own data
- Sensitive API integrations (Gmail, Calendar, Jira) use OAuth-based authentication
- Only required data is stored in the database, reducing exposure of sensitive information
- Input validation is performed before processing to prevent invalid or harmful inputs
- System logs are maintained to track activities and detect abnormal behaviour

-

6.4 Coding Standard

The following coding standards have been followed here:

- Class names are written using PascalCase (e.g., EmailSecurityAgent, SchedulerAgent)
- Function names follow a clear naming convention using descriptive verbs (e.g., processEmail(), generateSlides())
- Variables are named meaningfully to reflect their purpose (e.g., userInput, apiResponse)
- Code is organized into separate modules for:
 - Agents
 - API integrations
 - Database operations
- Reusable functions are created to avoid code duplication
- Proper indentation and formatting are maintained throughout the code
- Error handling is implemented using try-catch blocks or equivalent mechanisms
- Comments are added where necessary to explain complex logic
- API keys and sensitive data are stored securely using environment variables instead of hardcoding

CHAPTER 7: TESTING

7.1 TESTING PLAN

The testing of the CyberNexus system was performed in an incremental manner. After implementing each functionality (agent), testing was carried out immediately to verify its correctness before moving to the next module.

This approach helped in identifying issues at an early stage and ensured that each module was working properly before integration with the overall system.

7.2 TESTING STRATEGY

Testing was primarily performed by the developer during the development process. Since the same person was responsible for both development and testing, white-box testing was followed.

In addition:

- Unit testing was performed for each individual module (agent) after development
- Integration testing was performed when multiple agents and APIs were connected
- System testing was conducted after completing all functionalities to verify overall workflow

7.3 TEST SUITES DESIGN

Table 7.1 Authentication Module

Test Case ID	Input	Expected Output	Actual Output	Result
TS1-TC1	Valid email & password	Redirect to dashboard	Redirected successfully	Pass
TS1-TC2	Invalid email	Error message	Error message shown	Pass
TS1-TC3	Empty credentials	Validation error	Validation error shown	Pass

Table 7.2 AutoPresenter Agent

Test Case ID	Input	Expected Output	Actual Output	Result
TS2-TC1	Valid topic	PPT generated	PPT generated	Pass
TS2-TC2	Empty input	Error message	Error shown	Pass
TS2-TC3	JSON input	Structured slides	Slides generated	Pass

Table 7.3 DB Presenter Agent

Test Case ID	Input	Expected Output	Actual Output	Result
TS3-TC1	Natural language query	SQL generated	SQL generated	Pass
TS3-TC2	Valid query	Data fetched	Data displayed	Pass
TS3-TC3	Invalid query	Error message	Error shown	Pass

Table 7.4 Content Reviewer Agent

Test Case ID	Input	Expected Output	Actual Output	Result
TS4-TC1	Valid URL	Errors detected	Errors listed	Pass
TS4-TC2	Invalid URL	Error message	Error shown	Pass

Table 7.5 Header Validator Agent

Test Case ID	Input	Expected Output	Actual Output	Result
TS5-TC1	Valid URL	Header analysis	Report generated	Pass
TS5-TC2	Invalid URL	Error message	Error shown	Pass

Table 7.6 Resume Reviewer Agent

Test Case ID	Input	Expected Output	Actual Output	Result
TS6-TC1	Resume + JD	Score generated	Score displayed	Pass
TS6-TC2	Invalid file	Error message	Error shown	Pass

Table 7.7 Newsletter Bot

Test Case ID	Input	Expected Output	Actual Output	Result
TS7-TC1	Trigger execution	News fetched	News displayed	Pass
TS7-TC2	No data	Empty response handled	Handled correctly	Pass

Table 7.8 Compliance Agent

Test Case ID	Input	Expected Output	Actual Output	Result
TS8-TC1	Valid Jira details	Tickets fetched	Tickets displayed	Pass
TS8-TC2	Invalid credentials	Error message	Error shown	Pass

Table 7.9 Email Security Agent

Test Case ID	Input	Expected Output	Actual Output	Result
TS9-TC1	Phishing email	Classified as fraud	Correct classification	Pass
TS9-TC2	Normal email	Classified as safe	Correct classification	Pass

Table 7.10 Resume Reviewer Agent

Test Case ID	Input	Expected Output	Actual Output	Result
TS10-TC1	Available slots	Slots generated	Slots shown	Pass
TS10-TC2	Conflict case	Avoid overlap	No conflict	Pass

CHAPTER 8: CONCLUSION AND DISCUSSION

8.1 SELF ANALYSIS OF PROJECT VIABILITIES

The development of CyberNexus provided a strong understanding of how modern AI-based systems can be designed and implemented for real-world applications. During the project, I gained practical experience in working with AI frameworks, API integrations, and building a modular system using multiple agents.

One of the key strengths of this project is its multi-agent architecture, where different tasks are handled by specialized agents within a single platform. Unlike many existing systems that focus on a single domain, CyberNexus combines multiple functionalities such as cybersecurity analysis, recruitment automation, and content processing into one unified system.

The project also demonstrates the use of real-time integrations with external services like email systems, calendars, and project management tools, which makes it more practical and closer to real-world applications. Additionally, the implementation of authentication and multi-tenant architecture ensures that the system is secure and scalable.

From my perspective, the project is not only technically sound but also innovative, as it combines different domains into a single AI-driven platform. It highlights the potential of agent-based systems in improving efficiency and reducing manual effort.

8.2 PROBLEM ENCOUNTERED AND POSSIBLE SOLUTIONS

The following challenges were encountered during the development of the project:

- While integrating APIs such as Gmail, Jira, and VirusTotal, issues related to authentication, token expiration, and rate limits were faced.
- The AI model sometimes produced inconsistent or unexpected outputs, especially when input data was unclear or complex.

- Managing multiple agents and coordinating their workflows was challenging during initial development.
 - Implementing real-time features such as email monitoring required handling asynchronous events properly.
- Identifying errors in multi-layered systems (frontend, backend, AI, APIs) was complex.

The following possible solutions can be implemented:

- Proper error handling and token management mechanisms were implemented to ensure smooth API communication.
- Prompt design was improved, and input validation was added to ensure better and more consistent results.
- A centralized orchestrator and dispatcher mechanism was implemented to handle agent execution efficiently.
- Event-driven approaches and efficient API usage were applied to handle real-time workflows.
- Logging mechanisms and step-by-step debugging were used to trace and fix issues effectively.

8.3 SUMMARY OF PROJECT WORK

The CyberNexus project successfully demonstrates the design and implementation of an AI-powered multi-agent marketplace. The system integrates multiple functionalities into a single platform, enabling users to automate tasks related to cybersecurity, recruitment, and content analysis.

A total of nine different agents were developed, each handling a specific use case. The system supports secure authentication, real-time API integration, and intelligent processing using AI models. The modular architecture ensures scalability and allows easy addition of new features.

Overall, the project achieves its goal of reducing manual effort and improving efficiency through automation. It also provides a strong foundation for further enhancements and real-world deployment.

CHAPTER 9: LIMITATION AND FUTURE ENHANCEMENT

➤ **Limitations:**

- AI-generated outputs may not always be fully accurate and can vary based on input quality
- System performance depends on external APIs (Gmail, Jira, VirusTotal), and any downtime affects functionality
- API rate limits can restrict the number of requests processed in a given time
- Requires continuous internet connectivity for operation
- Real-time processing (e.g., email monitoring) may have slight delays due to API response time
- Limited handling of highly complex or unstructured input data
- UI is basic and can be further improved for better user experience

➤ **Future Enhancements:**

- Addition of more AI agents for extended functionalities
- Improvement in AI model accuracy using fine-tuning techniques
- Advanced analytics dashboard for better visualization of results
- Integration with more enterprise tools and platforms
- Enhancement of UI/UX for improved user interaction
- Deployment on scalable cloud infrastructure for handling larger user base

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